Lesson 1 | Print and set up

Print one teacher copy of the companion. For each class, print about 22 sets of individual handout PDF pages 2–5. Print group handout PDF page 2 once per pair or trio for Investigation Ideas. Adjust to actual enrollment. Students can use a notebook T-chart for Notice and Wonder.

Open the official Lesson 1 deck. Queue the 19-second Fallen Redwood Tree clip for slide 9. The embedded player appeared blank during preview. The YouTube copy has no captions. Use the visual transcript on Teacher Guide printed page 100 to prepare a description. Check classroom playback. If the clip is unavailable, use the five photos on slide 8 and record observations from the photos.

Set up four spaces for each class: Notice and Wonder, Initial Consensus Model, Related Phenomena, and the Driving Question Board. Set out markers, sticky notes, and six large sheets for groups of three or four.

Lesson 1 | Read this before teaching

Students will see plants and fungi growing on fallen trees. The images let them describe what they see and compare size and location. They cannot yet see what is inside the log or establish how a tree changes into a nurse log. The first model shows students' starting explanations. Keep uncertain ideas visible so Lesson 2 has questions to investigate.

What counts as progress today? Students point to something they observed, show a possible relationship in a model, and name a question a closer look could help answer.

Timing. Explore 30 minutes, Synthesize 30 minutes, Connect 20 minutes, Navigate 30 minutes. Use the publisher's sequence. When a block ends mid-activity, record the last completed slide and save the work named on that activity page.

Lesson 1 | Explore · Slides 2–13 · 30 minutes

1. Set up Notice and Wonder

Slide 2

Give students a T-chart. Ask them to write or sketch one thing they see and one thing they wonder. Invite labels in any language.

2. Look closely

Slide 3–7

Display the photographs one at a time. Ask, “Where do you see that?” On slide 6, ask how the people help students judge the size of the log.

3. Compare evidence

Slide 8–9

Use slide 8 to compare the five photos and slide 9 to observe the fallen redwood. Ask students what they noticed on this second look and what remains hard to see.

4. Share and record

Slide 10–12

Have groups rehearse one observation and one question. Record both on the class chart. If a student says, “The log feeds the plant,” ask, “What do you see, and what are you proposing might happen?”

5. Name the phenomenon

Slide 13

Introduce “nurse log” after students have shared their observations. Save the Notice and Wonder chart. Resume with slide 14 when Explore is complete.



Slide 8 photo collection. OpenSciEd Teacher Guide, printed p. 57.

Lesson 1 | Synthesize · Slides 14–19 · 30 minutes

1. Explain what a model does

Slide 14

Use the slide’s shadow example. Ask what the model helps explain.

2. Make an individual model

Slide 15

Use both Initial Model pages. Have students show what may happen on, under, and inside the log. Invite arrows and labels. Use question marks for uncertain ideas.

3. Compare with a partner

Slide 16–17

Gather in a Scientists Circle and revisit the class agreements. Have partners explain two similarities and two differences between their models.

4. Build a class model

Slide 18–19

Ask which ideas can go on the model now and which need evidence. If students disagree about whether roots reach through the log, record both possibilities. Ask what a closer observation could reveal.

5. **Save the work.** Keep the individual models as an ungraded baseline and preserve the class model. Resume at slide 20.

Lesson 1 | Connect and Navigate · Slides 20–25 · 50 minutes

Connect · 20 minutes

1. Invite a community example

Slide 20

Ask students to look for something that reminds them of the nurse log. They can talk with a family or community member and bring back an example if able.

2. Compare examples

Slide 21

Give groups large paper. Ask, “What is similar to the nurse log? What is different?” Record the connections students explain. Save the chart and resume at slide 22 if class ends here.

Navigate · 30 minutes

1. Write questions

Slide 22

Reopen the charts and model. Have students write one question per sticky note.

2. Build the Driving Question Board

Slide 23

Ask students to read their questions and explain which ones belong together. Keep distinct questions visible within each cluster.

3. Propose an investigation

Slide 24

Have pairs or trios choose a question and suggest something the class could observe. Ask how that observation would help.

4. Choose where to start

Slide 25

Use a student question to motivate taking a closer look at a nurse log in Lesson 2. Save each class’s board and model.

Lesson 1 | Understanding record · One per class

Class: _____ Date: _____

An observation students used as evidence

_____

A relationship students proposed in their models

_____

A question a closer look might answer

_____

Last completed slide and the first move next time

_____

Source: OpenSciEd Unit 5.1 Teacher Guide Version 1.0, printed pp. 50–73 and 100. Official deck slides A–X are deck positions 2–25. Print quantities use the estimated 22 students per class.